

Deflection of a Standard PLA Filament Bracket

TM

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1 Introduction

A 3D-printed bracket is made to support the load of a surround-sound speaker weighing $P = 6.35 \text{ lb}$. Given that the elastic bending modulus, E , of Ender PLA+ filament is 331701.906 psi , with each member having a 20% filled triangular lattice ($\delta_{20\%} \approx 1.31798830848\delta_{100\%}$), estimate the deflection, δ , of point B as shown in Fig. (1):

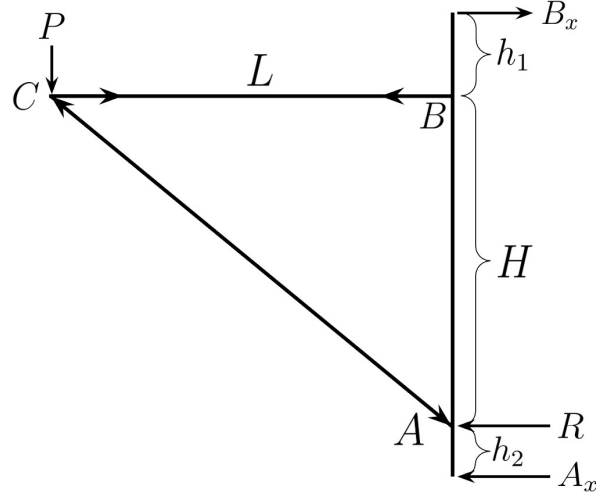


Figure 1: A triangular truss supporting a load, P , by two drywall screws at points A_x and B_x .

The following information is provided for Fig. (2):

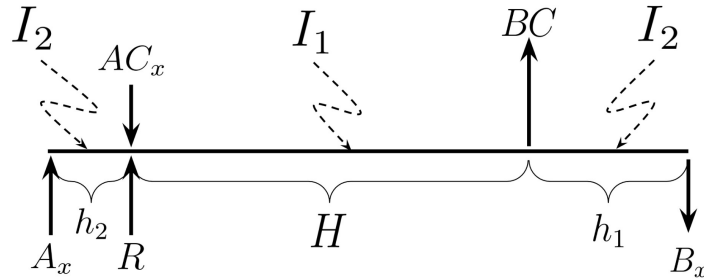


Figure 2: Member AB from Fig. 1. Assume that point A is cantilevered and point B is pinned (can freely rotate).

Givens:

- $P = 6.35 \text{ lb}$
- $E = 331701.906 \text{ psi}$

- $H = 4 \text{ in.}$
- $L = 5.5 \text{ in.}$
- $h_1 = 1.5 \text{ in.}$
- $h_2 = 1.0 \text{ in.}$
- $I_1 = 0.004 \text{ in}^4$ for $h_2 \leq x \leq H + h_2$
- $I_2 = 0.0005 \text{ in}^4$ for $0 \leq x \leq h_2$ and $H + h_2 \leq x \leq H + h_1 + h_2$

2 Solution

A quick statics analysis by the method of joints yields,

$$AC = \frac{P\sqrt{L^2 + H^2}}{H}$$

$$AC_x = \frac{PL}{H}$$

$$BC = \frac{PL}{H}$$

$$A_x = \frac{PL}{H + h_1 + h_2}$$

the top screw's resistance, B_x , can be determined.

$$\circlearrowleft \sum_i^n M_A = 0 = PL - B_x (H + h_1 + h_2)$$

$$B_x = \frac{PL}{H + h_1 + h_2}$$

We invoke the method of superposition on member AB . Because the diagonal brace acts against the wall, the deflection between $x = 0$ and $x = h_2$ is negligible. By letting point A be a cantilever, we have a statically indeterminate system in which the reaction force, R , is unknown. Negating h_2 , let $x = 0$ be the location of R .

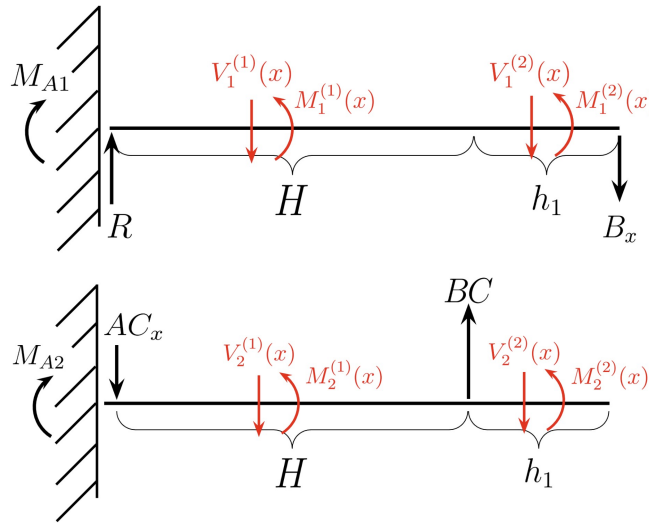


Figure 3: Two separated problems each with their shear and moment functions denoted by the subscript. The superscript indicates the section pertaining to the inertias, I_1 and I_2 .

The deflection of the n th interval of x is,

$$\frac{d^2\delta_i}{dx^2} = \frac{d\theta_i}{dx} = \frac{S \cdot M^{(i)}(x)}{EI_i}$$

where E is the modulus of elasticity of the PLA plastic, I_i is the moment of inertia of the i th section, and S is a factor pertaining to the percent PLA filler in the structural member. Experimentally, for 20% filler of a triangular lattice, it was determined that $S \approx 1.339869$.

The deflection of the member AB given different inertia, I_n , in each interval x distance along it is an iterative process: determine the bending moment functions, solve the second-order deflection differential equations, and solve the coefficients given some assumed boundary conditions.

2.1 Case 1: Downward Deflection

We begin by determining the reacting moment at point A and force, B_x :

$$\begin{aligned} \uparrow^+ \sum F_x &= 0 = R - B_x \\ \therefore B_x &= R \end{aligned}$$

$$\begin{aligned} \curvearrowright^+ \sum M_A &= 0 = -M_{A1} - B_x(H + h_1) \\ \therefore M_{A1} &= -R(H + h_1) \end{aligned}$$

For $0 \leq x \leq H$, the moment function along AB is determined to be

$$\begin{aligned} \curvearrowright^+ \sum M &= 0 = M_1^{(1)}(x) - Rx - M_{A1} \\ \therefore M_1(x)^{(1)} &= M_1^{(2)}(x) = Rx - R(H + h_1) \end{aligned}$$

Given the boundary conditions, $\theta_1^{(1)}(0) = 0$, $\delta_1^{(1)}(0) = 0$,

$$\begin{aligned} \theta_1^{(1)}(x) &= \frac{S}{EI_1} \int M_1(x) dx \\ &= \frac{S}{EI_1} \int (Rx - R(H + h_1)) dx \\ &= \frac{S}{EI_1} \left[\frac{Rx^2}{2} - R(H + h_1)x + a_1 \right] \quad , \quad \theta_1^{(1)}(0) = 0 \Rightarrow a_1 = 0 \\ \delta_1^{(1)}(x) &= \frac{S}{EI_1} \left[\frac{Rx^3}{6} - \frac{Rx^2}{2}(H + h_1) + a_2 \right] \quad , \quad \delta_1^{(1)}(0) = 0 \Rightarrow a_2 = 0 \\ \therefore \delta_1^{(1)}(x) &= \frac{SR}{EI_1} \left[\frac{x^3}{6} - \frac{x^2}{2}(H + h_1) \right] \end{aligned} \tag{1}$$

And,

$$\begin{aligned} \theta_1^{(2)}(x) &= \frac{S}{EI_2} \int (Rx - R(H + h_1)) dx \\ &= \frac{S}{EI_2} \left[\frac{Rx^2}{2} - R(H + h_1)x + b_1 \right] \\ \theta_1^{(2)}(H) &= \theta_1^{(1)}(H) \Rightarrow b_1 = \left(-\frac{RH^2}{2} + RH(H + h_1) \right) \left(1 - \frac{I_2}{I_1} \right) \\ \delta_1^{(2)}(x) &= \frac{S}{EI_2} \left[\frac{Rx^3}{6} - \frac{R(H + h_1)x^2}{2} + \left(-\frac{RH^2}{2} + RH(H + h_1) \right) \left(1 - \frac{I_2}{I_1} \right) x + b_2 \right] \\ \delta_1^{(1)}(H) &= \delta_1^{(2)}(H) \Rightarrow b_2 = \left(\frac{RH^3}{3} - \frac{RH^2(H + h_1)}{2} \right) \left(1 - \frac{I_2}{I_1} \right) \\ \therefore \delta_1^{(2)}(x) &= \frac{SR}{EI_2} \left[\frac{x^3}{6} - \frac{(H + h_1)x^2}{2} + \left(\frac{H^2}{2} + Hh_1 \right) \left(1 - \frac{I_2}{I_1} \right) x - \left(\frac{H^3}{6} + \frac{H^2h_1}{2} \right) \left(1 - \frac{I_2}{I_1} \right) \right] \end{aligned} \tag{2}$$

2.2 Case 2: Upward Deflection

The reaction forces and moments on AB for case 2 is,

$$\begin{aligned}\circlearrowleft^+ \sum M_A &= 0 = -M_{A2} + BCH \\ \therefore M_{A2} &= BCH\end{aligned}$$

$$\begin{aligned}\uparrow^+ \sum F_x &= 0 = BC - AC_x \\ \therefore AC_x &= BC\end{aligned}$$

For $0 \leq x \leq H$

$$\begin{aligned}\circlearrowleft^+ \sum M &= 0 = M_2^{(1)}(x) + AC_x x - M_{A2} \\ \therefore M_2^{(1)}(x) &= BC(H - x)\end{aligned}$$

For $H \leq x \leq H + h_1$

$$\begin{aligned}\circlearrowleft^+ \sum M &= 0 = M_2^{(2)}(x) + AC_x x - M_{A2} - BC(x - H) \\ \therefore M_2^{(2)}(x) &= 0\end{aligned}$$

Given the following boundary conditions, we have,

$$\begin{aligned}\theta_2^{(1)}(x) &= \frac{S}{EI_1} \int M_2^{(1)}(x) dx \\ &= \frac{S}{EI_1} \int BC(H - x) dx \\ &= \frac{S}{EI_1} \left[BC \left(-\frac{x^3}{6} + \frac{Hx^2}{2} \right) + c_1 \right] \quad , \quad \theta_2^{(1)}(0) = 0 \Rightarrow c_1 = 0 \\ \delta_2^{(1)}(x) &= \frac{S}{EI_1} \left[BC \left(-\frac{x^3}{6} + \frac{Hx^2}{2} \right) + c_2 \right] \quad , \quad \delta_2^{(1)}(0) = 0 \Rightarrow c_2 = 0 \\ \therefore \delta_2^{(1)}(x) &= \frac{SBC}{EI_1} \left(-\frac{x^3}{6} + \frac{Hx^2}{2} \right)\end{aligned} \tag{3}$$

And,

$$\begin{aligned}\theta_2^{(2)}(x) &= \frac{S}{EI_2} \int M_2^{(2)}(x) dx \\ &= \frac{S}{EI_2} \int 0 dx \\ &= \frac{S}{EI_2} d_1 \quad , \quad \theta_2^{(2)}(H) = \theta_2^{(1)}(H) \Rightarrow d_1 = BC \frac{H^2}{2} \frac{I_2}{I_1} \\ \delta_2^{(2)}(x) &= \frac{S}{EI_2} \left[BC \frac{H^2}{2} \frac{I_2}{I_1} x + d_2 \right] \quad , \quad \delta_2^{(2)}(H) = \delta_2^{(1)}(H) \Rightarrow d_2 = -BC \frac{H^3}{6} \frac{I_2}{I_1} \\ \therefore \delta_2^{(2)}(x) &= \frac{SBC}{EI_1} \left(\frac{H^2}{2} x - \frac{H^3}{6} \right)\end{aligned} \tag{4}$$

2.3 Solving for the Reaction, R :

We seek a solution to R in EquationS (1) and (2) in the upward deflection such that,

$$\delta_1^2(H + h_1) + \delta_2^2(H + h_1) = 0$$

Working through the algebra, we obtain,

$$\begin{aligned} \frac{SBC}{EI_1} \left(\frac{H^2(H+h_1)}{2} - \frac{H^3}{6} \right) &= -\frac{S}{EI_2} \left[\frac{R(H+h_1)^3}{6} - \frac{R(H+h_1)^3}{2} + \right. \\ &\quad \left. \left(\frac{RH^2}{2} + RHh_1 \right) \left(1 - \frac{I_2}{I_1} \right) (H+h_1) - \left(\frac{RH^3}{6} + \frac{RH^2h_1}{2} \right) \left(1 - \frac{I_2}{I_1} \right) \right] \\ -BC \frac{I_2}{I_1} \left(\frac{H^2(H+h_1)}{2} - \frac{H^3}{6} \right) &= R \left[-\frac{(H+h_1)^3}{3} + \left(\frac{H^2(H+h_1)}{2} + Hh_1(H+h_1) - \frac{H^3}{6} - \frac{H^2h_1}{2} \right) \left(1 - \frac{I_2}{I_1} \right) \right] \end{aligned}$$

and with $BC = \frac{PL}{H}$,

$$R = \frac{PLI_2}{I_1} \left(\frac{H(H+h_1)}{2} - \frac{H^2}{6} \right) \left[\frac{(H+h_1)^3}{3} + \left(1 - \frac{I_2}{I_1} \right) \left(H^2(H+h_1) - H(H+h_1)^2 - \frac{H^3}{3} \right) \right]^{-1}$$

3 Calculated vs. Experimental Results:

We may obtain a flexural modulus from Table 3 of (Caminero 2019) [1]. Because the bracket was printed with the triangle on its side, the "flat" orientation's flexural modulus is $E_f = 2.287 \text{ GPa} = 331,701.906 \text{ psi}$.

Four deflection tests were performed on beams with 20% infill, each having unique cross-sectional bases b , heights h , and lengths l . A 20lb dumbbell was placed on each beam's midpoint while supported at their ends, where deflection, $\delta_{20\%}$, was measured using a caliber.

The following unity ratios in Table (1) were obtained using Equation (5).

$$\delta_{100\%} = \frac{Pl^3}{48EI} \quad (5)$$

$b \text{ (in.)}$	$h \text{ (in.)}$	$l \text{ (in.)}$	$I_x \text{ (in.}^4\text{)}$	$\delta_{100\%} \text{ (in.)}$	$\delta_{20\%} \text{ (in.)}$	$\delta_{100\%}/\delta_{20\%}$
0.20	0.28	3.80	0.000365867	0.188	0.225	0.83731
0.46	0.28	4.65	0.000841493	0.150	0.240	0.62537
0.39	0.28	2.80	0.000713440	0.039	0.045	0.85891
0.97	0.46	5.88	0.000786799	0.032	0.046	0.71334

Table 1: Base b , height h , inertias I , calculated deflections $\delta_{100\%}$, measured deflections $\delta_{20\%}$, and unity ratios, $1/S_i$.

A bending increase factor may be found by,

$$\begin{aligned} S &= \frac{1}{n} \sum_{i=1}^n \left(\frac{\delta_{20\%}}{\delta_{100\%}} \right)_i \\ &= \frac{1}{4} [1.194 + 1.599 + 1.164 + 1.402] \\ &= 1.33986919414 \end{aligned} \quad (6)$$

Applying the 20 lb dumbbell directly on point B of member AB of the bracket, the measured deflection was approximately 0.13 in. Letting $BC = 20 \text{ lb}$ (or $P = 14.545 \text{ lb}$) and $S = 1.339 \dots$ in,

$$\delta^{(1)}(x) = \delta_1^{(1)}(x) + \delta_2^{(1)}(x) \quad (7)$$

$$\delta^{(2)}(x) = \delta_1^{(2)}(x) + \delta_2^{(2)}(x) \quad (8)$$

and finding the global maximum of either of these equations, the predicted deflection is $\delta_{20\%} = 0.1061 \text{ in.}$ This yields a unity ratio of $\delta_{20\%}/\delta_{measured} = 0.8164$ due to the high deflection of the h_1 region and the lack of a cantilevered support at h_2 .

Equations (7) and (8) predict that the 6.35 lb speaker will cause a deflection of $\delta_{20\%} = 0.033698 \text{ in.}$

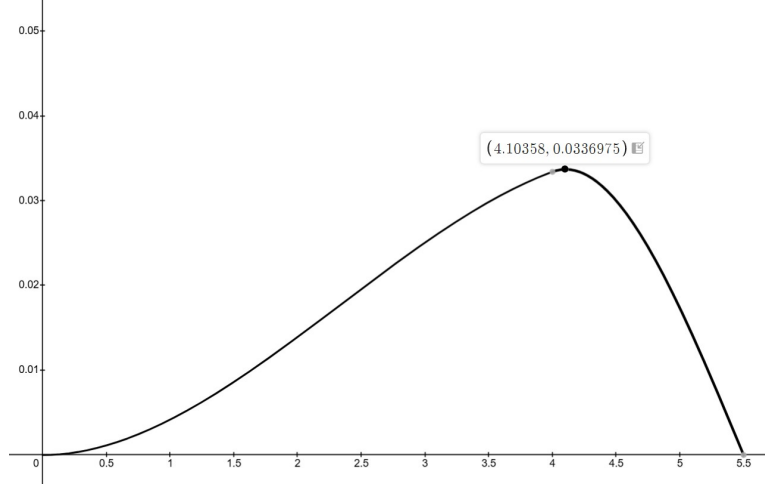


Figure 4: Predicted deflection due to the speaker along a vertical distance, x , along member AB .

References

- [1] E. A. MIGUEL CAMINERO, *Additive manufacturing of pla-based composites using fused filament fabrication: Effect of graphene nanoplatelet reinforcement on mechanical properties, dimensional accuracy and texture*, MDPI, (2019).